How to Interview

The interview provides the opportunity for the finalist to tell you what they know, and answer questions, so that you as a judge can determine the quality of their research and the quality of the work relative to the other projects within the category. Too often, the display does not appropriately capture the accomplishments of outstanding projects and the talent of the students who developed them. In the end, the interview is the critical element. The display is secondary to the finalist's actual knowledge of the subject and explanation of the work.

Spirit of Interviewing

While interviewing students:

- Be friendly, courteous, and professional.
- Demonstrate an interest in what the finalist has done.
- Employ active listening.
- Attempt to learn something from each project.
- Differentiate nervousness/shyness from lack of knowledge.

Also, please read Document *D: The Spirit of ISEF Judging*, in Section I of the *Judge Information Board*.

Interview Timing

At ISEF, a new interview starts every 15 minutes: on the hour, and at 15, 30, and 45 minutes after the hour. The interview should start as soon as you arrive at the display and greet the student. You have only 10-12 minutes for the actual formal interview. The remainder of the 15-minute period will be taken up by introductions at the beginning, extricating yourself from the interview at the end, recording notes, and then moving to the next project.

An auditory signal will be broadcast throughout the project display area at 12 minutes into each interview session. This is your signal to wrap-up of the interview. Every 15 minutes there will be a verbal announcement indicating the beginning of the next interview period. Before that announcement, you should already be at your next interview.

Judges should always be where they are scheduled to be, when they are scheduled to be there. Do not deviate from your printed schedule. This causes problems for the student you are interviewing, for the student you should be interviewing, and for the other judges. The student is entitled to your full attention when you are scheduled to meet them. Don't shortchange them. Don't come late or leave early.

If there are unresolved or overlooked questions after an interview, do not go back to the student. These questions can be resolved in conversation with other judges or brought up at the caucus. Other judges, who have more familiarity with the applicable science, will have spoken to the student and can help sort it out.

Starting the Interview

The interview should start with the finalist explaining their project. All finalists have a prepared speech. Typically, these speeches are too long for your purposes. Some students would talk on their own for the entire time if allowed. It is the duty of the judge to take control of the interview and not allow the finalist to monopolize the time with their prepared speech. After all, you have a very limited time to spend with the student, so you cannot afford for them to recite a canned speech covering issues that you already understand.

Many judges choose to allow either a short introduction (1-2 minutes) or something longer (3-5 minutes). It is fair for you to instruct the finalist to begin their prepared presentation, but warn them that you may interrupt with questions as they arise. Make sure that the finalist knows what your expectations are.

Often students are at one of two extremes: some who are shy and some who talk too much. For those who are shy, ask questions to get them started. Try to delve into something personal. Why did you pick this topic? What is your interest in this category? In the other direction, you can stop students from talking too much by interrupting to ask questions.

Ideally, you will have some specific questions that you formulated during the review of the materials before the interview day. This way, you will demonstrate respect for the student showing that you have reviewed their project and are prepared to conduct the interview. These questions should focus on:

- What is new/unique about this project?
- How did the student collect data and control data quality?
- Does the student understand the effect of one variable on others?
- What work did the student do, and what was done for the student?
- Does the student know the science/engineering principles behind the project?
- Did the student design the project or was it given to them, perhaps as part of a larger research effort?
- What might the student do differently the next time around?

Judges should not be afraid to ask difficult or probing questions, but it is important that the judge be aware of the tone of voice used to ask the question and their body posture during the questioning. Be careful not to intimidate students. Similarly, do not convey your actual decision regarding the project, whether that be positive or negative.

Professional Research Assistance

Many finalists have done their work in university labs or other professional environment. In this case, there may be some difficulty in determining what the student did versus what was done for them. Judges must determine the degree of independence of the finalist in conducting the project through questions and reviewing ISEF Form 1C, *Regulated Research Institutional/Industrial Setting Form*. All students can get help. It is the judge's judgment on what is appropriate or inappropriate. Appropriate help would be providing guidance, materials and logistical support that the student needs in order to complete their project. Giving a student a “cookbook” project is not appropriate. Students would not be expected to use power tools or learn trade skills such as welding. If they do, it may be a plus, but not doing it is never a minus. In professional research, running lab tests is often done by others, especially where the instruments are valuable or difficult to use. If the students learn to perform lab tests themselves, then this is a plus. A scientist may study coronavirus and collect thousands of samples in the course of research, but never do any tests themselves. These all go to the virology lab. Judge students accordingly.

A simple way to evaluate the value of the help is to assess the depth of the student’s fundamental understanding. If they understand the underlying science, then the level of help was probably appropriate. Here are some sample approaches you could take:

- Ask the student to explain the fundamental science.
- Ask the student to explain how they isolated their data to just one affected variable.
- Ask the student directly what help they received and why they needed that precise help.
- Ask how their conclusion/apparatus would have changed without outside help.

Team Projects

Team projects are those conducted by 2 or 3 students working together. At ISEF, team and individual projects are judged together. Projects should be judged only on the basis of their quality. However, all team members should demonstrate significant contributions. Determining this can be as simple as asking who did what. All members should have an understanding of the underlying science/engineering principles, and an understanding of the project as a whole. Failure to have a true team approach would weigh against the final score of the project.

Apparently Futile Interviews

There are some situations where you may have no good questions for the student:

1. The Project is Fatally Flawed

Every student deserves your time when you are scheduled to interview him or her. If you perceive that a project is so poorly designed and executed it has no chance of winning an award, you must still provide a full interview. Ask questions that will give the student the opportunity to explain his or her work. If, after giving the student sufficient opportunity, they have only convinced you further that the project is not worthy of an award, you could respond by giving the student additional time to present more of their canned speech. See also, the bulleted list of general questions below.

2. You Don’t Understand the Project

On occasion, any judge can come across a project that they do not understand, perhaps because it is too far outside their area of expertise, or perhaps because the student explained their project so badly as to be unrecognizable. This can be frustrating, and even intimidating if you are facing an incredibly impressive display and a supremely confident student. In this case, ask questions on the basic science or engineering and the scientific method. Again, see below.

3. The Student Doesn't Understand the Project

Sometimes students do projects in technical areas with which a judge is intimately familiar, and the student just didn't get it — they made some incorrect assumptions, missed a key indicator in the data, came up with a false conclusion, or didn't look at or understand some common principles. It can be tempting to share knowledge about the topic, to help the student appreciate what happened (or should have happened) in the experiment. **Do not provide this help during the interview.** Help provided during the interview will resurface with subsequent judges under the guise of the student's own hard work. If you want to help them after the fair, see Document *H: Ethical Concerns for Judges* in Section II for the proper protocol.

Students sometimes do not understand what they have actually done and just make stuff up. In this instance, ask for explanations of words that the student used. Do not assume the student knows what each technical term means. They may recognize the generic term but be unable to explain what it means in the current context. Concerning equipment they have used, they may not know what the specific piece of equipment does, how it works, or even why it was used.

Ask the student to explain the fundamental science. Some examples might be, "How does a semiconductor work?" "How does an airplane fly?" "What is photosynthesis?" A student who was been helped too much by others often cannot explain the basic concepts. If the student cannot explain the basic science or engineering clearly, then the student really doesn't understand the science of what's going on. Chances are, if it doesn't make sense to you, it doesn't make sense. If you are still not certain, discuss the issue with other judges in your caucus room; there will probably be other judges with similar reservations.

Suggested General Questions

In each of the above apparently futile situations, you could start by letting the student go on with their canned speech a little longer than usual. You could have the student explain the underlying science. Here are some very general questions you can ask of any project.

- How did you come up with the idea for this project?
- What did you learn from your background search?
- How long did it take you to build the apparatus?
- How did you build the apparatus?
- How much time (how many days) did it take to run the experiments (grow the plants) (collect each data point)?
- How many times did you run the experiment with each configuration?
- How many experiment runs does each data point on the chart represent?
- Did you take all data (run the experiment) under the same conditions, *e.g.*, at the same temperature (time of day) (lighting conditions)?
- How does your apparatus (equipment) (instrument) work?
- What do you mean by (terminology or jargon used by the student)?
- What are the applications in industry for this knowledge or technique?
- Were there any books that helped you do your analysis (build your apparatus)?

- When did you start this project?
- What is the next experiment to do in continuing this study?
- Are there any areas that we not have covered which you feel are important?

Not all of the above questions are of equal value. For example, the first question above, regarding the reason the student chose their project, is friendly, but not particularly relevant to the actual research. For example, if the student's grandmother has liver cancer and that is why they chose to study liver cancer, this has no bearing on the quality of the work. The vignette is heartwarming, but not useful for judging purposes.

Always ask questions to learn more about the project. Never ask questions to embarrass or intimidate the student. This sounds simple but can be challenging to implement. Never cut the interview short. Each student deserves a full interview period with the judge.

Do Not Provide Feedback to the Students

Judges do not provide feedback to the finalist about their project. Ideally, the student should have no idea of the judge's evaluation of the project. Avoid suggesting corrections, possible extensions, or any other feedback that the student can bring to other judges. However, no matter how impressive or unimpressive the project, it is always appropriate for a judge at the end of the interview to recognize or even praise the student's accomplishment as an ISEF finalist and encourage their further work in their field.

Judges should never give students the answers to the judge's questions. These answers will distort the judging process by subsequently letting them give those answers to other judges. Similarly, do not prejudice other judges by telling them ahead of their interviews that the finalist is wrong, confused, etc. Students sometimes take a judge's reaction to determine what is good or not good about their project, and how they are doing toward receiving an award. Students will use any information they can infer from you with other judges. Do not provide more support for the old science fair adage, "students get smarter as the day progresses."

Special Circumstances

Finally, you may come across some projects with the following special circumstances

Interpreter is Present

English is the official language of ISEF. However, some students with limited or no ability to speak English will need to have an interpreter at their interview, or will have requested an interpreter to provide a resource should communicating with judges be difficult. Please see Document *E: Interviewing Students Who Don't Speak English* in Section II of the *Judge Information Board* for guidance.

Problems Outside of Science or Engineering

The job of a judge is to consider only the scientific/engineering merit of the student project. Any other problem with the student (*i.e.*, suspected plagiarism, inappropriate dress, insulting comments, etc.) should not affect the scoring of the work. Instead, take your concern to your category co-chairs. Do not communicate these problems to the student or to other judges.

Instead, please see Document *G: Student Scientific Misconduct* in Section II of the *Judge Information Board*.

Multi-Year Projects

Not infrequently, you will come across projects labeled "Phase 2" or "Year 3" or are otherwise obviously multi-year efforts, generically called "continuation projects." Such projects will have had to submit the previous year's abstract, the previous year's research plan or project summary, and a completed ISEF Form 7, *Continuation/ Research Progression Projects Form*. The student is also required to post these documents to the project display before ISEF would approve the project for judging.

Judges must evaluate ONLY work done in the "current year," defined by ISEF as any continuous 12-month period starting no earlier than January of the previous calendar year. When you encounter a continuation project, use ISEF Form 7 to clarify what the student did in the "current year" and how it differs from work that they did prior to the "current year."

Points of Emphasis

1. Treat all students with respect.
2. You have only 10-12 minutes for the actual interview with the student.
3. *Stick to your schedule.* Recognize the 12- and 15-minute signals for what they are.
4. Do not provide answers to students that could affect future interviews.
5. Never tell a student that you think their project is or is not award worthy.